04 Sorting and Aggregate Functions

Create Database:

```
-- Create database
CREATE DATABASE ecommerce;
USE ecommerce;
```

Create Tables:

```
-- Create Employees table
CREATE TABLE employees (
    empid INT PRIMARY KEY AUTO_INCREMENT,empname VARCHAR(100) NOT NULL,
    gender VARCHAR(10), age INT, departmentid INT,
    salary DECIMAL(10,2), hire_date DATE DEFAULT (CURRENT_DATE)
);

CREATE TABLE department (
    depid INT PRIMARY KEY AUTO_INCREMENT,name VARCHAR(100) NOT NULL
);

CREATE TABLE product (
    prodid INT PRIMARY KEY AUTO_INCREMENT,name VARCHAR(100) NOT NULL,
    price DECIMAL(10,2),stock_qty INT DEFAULT 0
);

CREATE TABLE sales (
    salesid INT PRIMARY KEY AUTO_INCREMENT,prodid INT,
    qty INT,sale_date DATE DEFAULT (CURRENT_DATE),
    FOREIGN KEY (prodid) REFERENCES product(prodid)
);
```

```
SHOW TABLES;
```

Tables_in_ecommerce
department
employees
product
sales

1: ORDER BY & LIMIT:

```
SELECT * FROM Product
ORDER BY PRICE desc LIMIT 3;
```

prodid	name	price	stock_qty
4	Laptop	55000.00	20
12	Television	40000.00	10
13	Air Conditioner	35000.00	12

2: Aggregate Functions:

```
SELECT COUNT(*) AS Tot_Sales FROM Sales;
SELECT SUM(qty) AS Tot_Qty FROM Sales;
```

Tot_Qty
42

Tot_Sales
13

```
SELECT AVG(price) AS Avg_Price FROM Product;
SELECT MAX(price) AS Max_Price FROM Product;
SELECT MIN(price) AS Min_Price FROM Product;
```

Avg_Price
16669.230769

Max_Price
55000.00

Min_Price
800.00

3: GROUP BY & HAVING:

```
SELECT DEPARTMENTID,count(*) AS Tot_Employees FROM Employees
GROUP BY DEPARTMENTID HAVING Tot_Employees > 3;
```

DEPARTMENTID	Tot_Employees
1	5
2	4